SMRG Flusim Web Dashboard
User Manual

This document will teach you how to use the SMRG Flusim Web Dashboard to run and visualise infectious disease simulations.

Contents

1 Configuring Simulations

1.1 Setting Baseline Parameters

The SMRG Flusim Web Dashboard allows for many aspects of a simulation to be configured. In order to run a simulation, you must first set these parameters to values matching the disease you wish to simulate. Click on the " Baseline Parameters" link on the sidebar (see Figure 1) to view the parameters that can be changed.

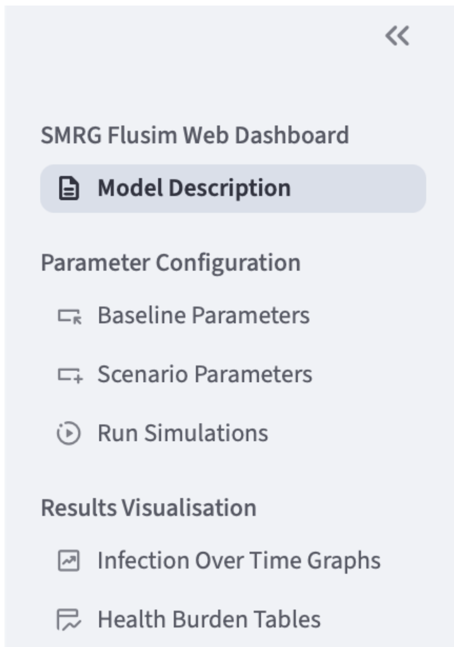

Figure 1: The navigation sidebar. Click on one of the page names to visit that page. If you can't see this sidebar, click the » arrow on the top-left of the page to make it visible.

The parameters on this page are split into 4 tabs based on what aspect of the simulation they modify, as seen in Figure 2. Click on any of these tabs to view its parameters.

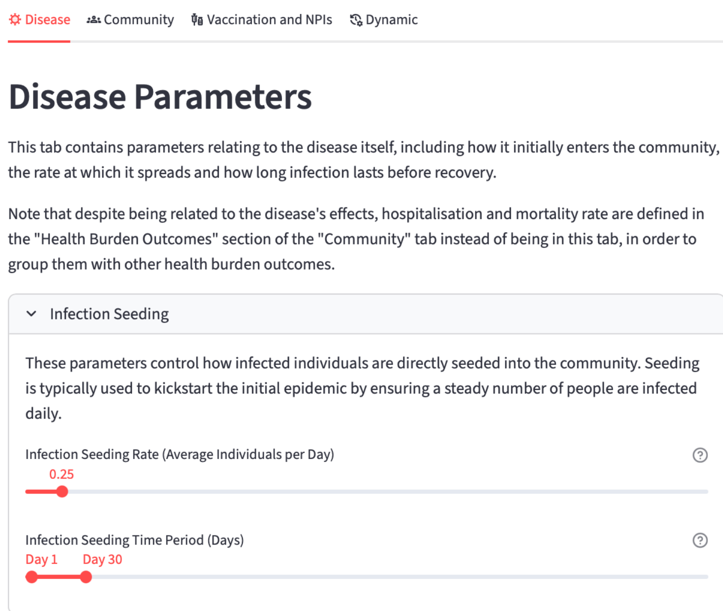

Figure 2: The Disease tab, with a pair of sliders controlling the rate at which infected individuals are seeded into the simulation. Click on one of the tab names at the top to switch to that tab. Click and drag the circles on the sliders to change the infection seeding settings.

Disease relates to the infection itself, including how frequently it infects new individuals and the length of time an infection lasts for.

Community controls the simulated population and how they react to the disease, including the size of gatherings and how likely a sick individual is to stay at home.

Vaccinations and NPIs controls vaccination strategies and non-pharmaceutical interventions (techniques used to limit the spread of disease without medication, e.g. social distancing).

Dynamic allows factors such as NPI effectiveness to change at specific times.

In each tab, there will be several parameters that can be adjusted. Most parameters use sliders for their input, as seen in Figure 2. Click and drag the slider to change the value of its parameter. If you do not know what a specific parameter does, hover your mouse over the ? symbol at the top-right of that parameter to show a description. Some parameters are enclosed in drop-down boxes like the one in Figure 2; click on a box to open or close it.

1.2 Parameter Types

There are several different ways that parameters on the dashboard are changed. Below is a list of the different input methods.

Sliders

Sliders are the most common method of setting parameters on the dashboard. In addition to clicking and dragging the slider, you can also click a point on the bar to move the slider there immediately, or use the left and right arrow keys to fine-tune the values of sliders.

Some sliders, like the one in Figure 3, have two values on the same line. The two values are the start and end of the range the parameter uses; either one can be adjusted freely.

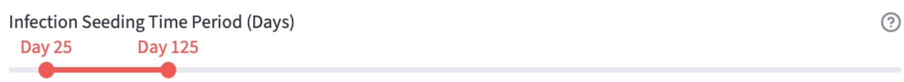

Figure 3: A slider with two values to adjust. This selects a period of time over which infected people will be added to the simulation.

Toggles

Some parameters can either be enabled or disabled in the simulation. Parameters like the one shown in Figure 4 can be toggled on and off by clicking the toggle switch. Note that some parameters are dependent on these toggles; for example, if vaccines are toggled off you won't be able to edit other vaccine-related parameters unless you toggle them back on.

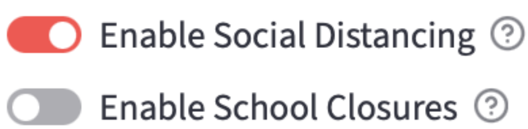

Figure 4: Toggle parameters which control whether specific non-pharmaceutical interventions are utilised in the simulation. The top one is enabled, while the bottom one is disabled.

Selections

If there are multiple options for a parameter, it will use a selection box like the one in Figure 5. Click on the box to reveal the options, then click on the option you want. Remember that hovering your mouse over the ⓘ symbol will show a description of the parameter, including what each option relates to.

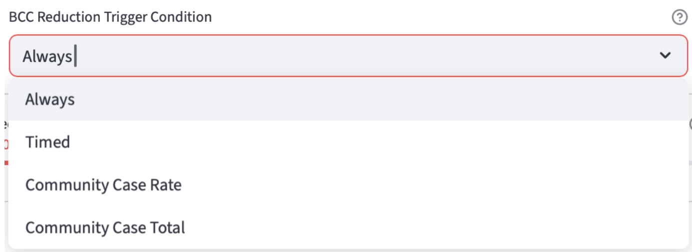

Figure 5: A parameter that uses a selection box to determine what type of condition will enable the background contact count reduction NPI in the simulation.

Number Inputs

Parameters like Figure 6 require you to enter a number. Simply type the number you want and press Enter to confirm, or click the **+** and **−** symbols to increase or decrease the value by a small amount. If you enter an invalid value (e.g. a negative number), the dashboard will prompt you to enter something else.

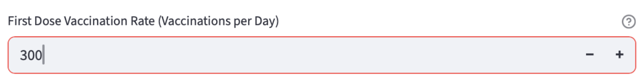

Figure 6: A number input parameter controlling how many people receive their first vaccine every day.

Tables

Some parameters can be set to a unique value for different age groups, such as the one in Figure 7. For parameters like these, a table is used to set the values for each age group. Double-click on a cell in the table to begin editing its value. For "Age Group" columns, select the age group that you would like to define age-specific parameter for using the drop-down menu. For numeric columns, simply type the new value and press Enter on your keyboard; if the column uses percentages, you will need to enter the value as a decimal.

You may add or remove rows from these tables freely. To add a row, hover

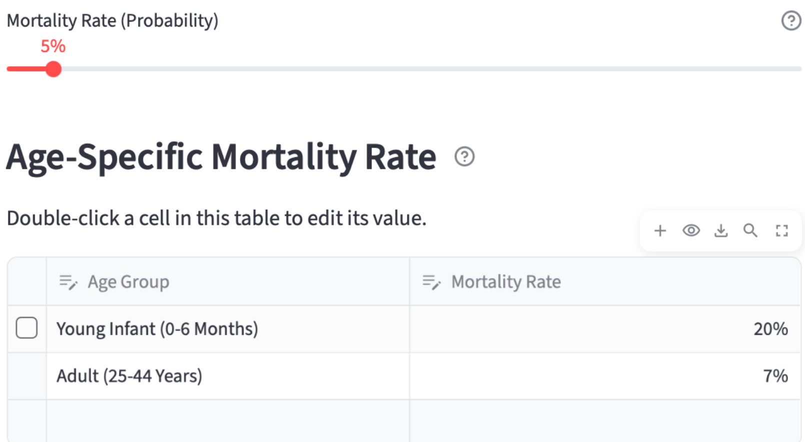

Figure 7: A parameter that uses a table to give different age groups different likelihoods of dying in the simulation.

your mouse over the row at the bottom of the table until a + icon appears, then click to add the row. Make sure to select an age group once the row has been added. Each row should use a different age group; if multiple rows have the same age group an error message will appear (see Parameter Errors for more details). To remove a row, click on the ☐ check box next to the row in the thin column to the left of the table, then click the delete button in the top-right corner of the table. Note that these buttons will only be visible when your mouse cursor is hovering over their location.

The parameters in the Dynamic tab also use tables for input. Instead of selecting an age group with the left column, you may enter a numbered day of the simulation where a parameter's value will change.

1.3 Additional Scenarios

In order to compare the effects that different parameters have on the simulation, the dashboard allows you to define multiple groups of parameter settings, or 'scenarios', at once.

The parameters set on the " Baseline Parameters" page are the values that will be used by the first scenario, also called the baseline scenario. In order to add additional scenarios, use the sidebar to navigate to the " Scenario Parameters" page. Click the "+ Add Scenario" button to add an additional scenario to the dashboard.

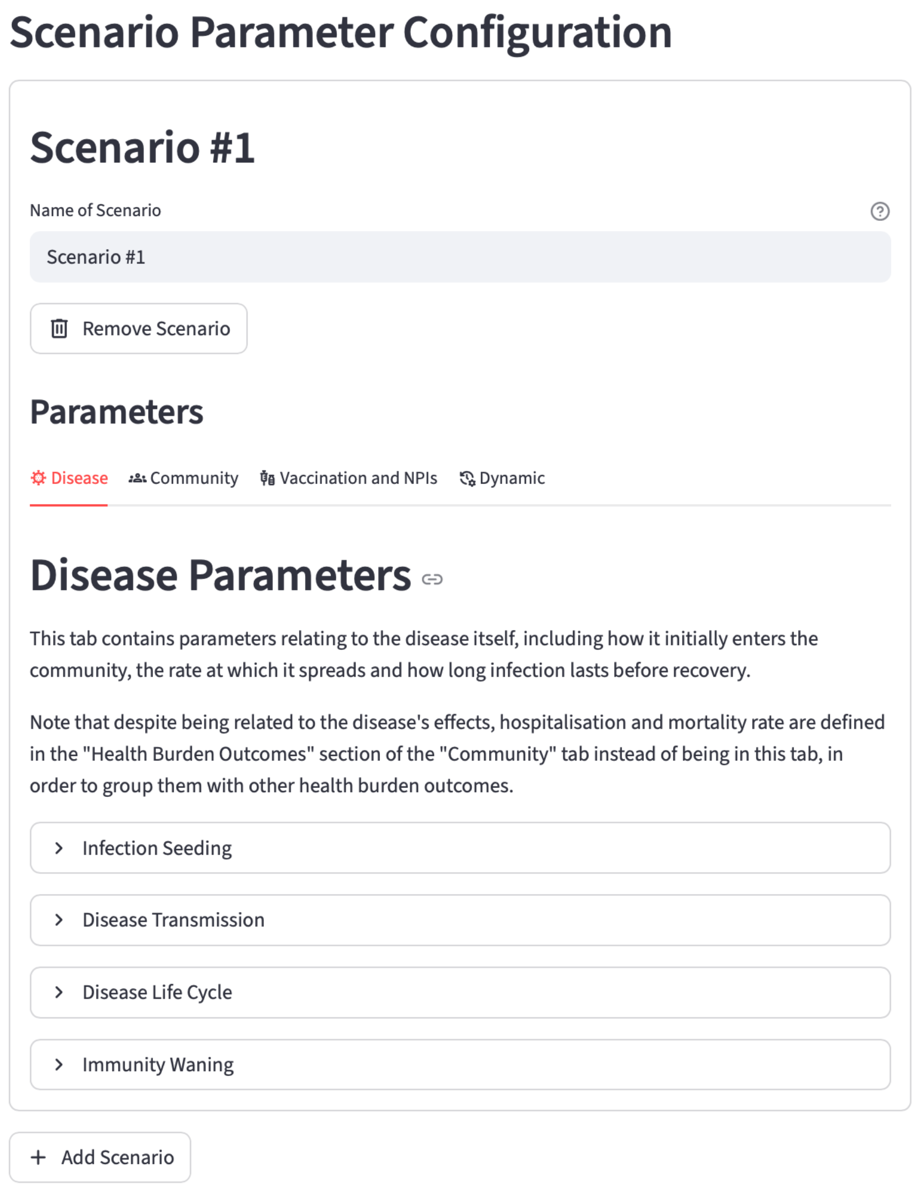

Figure 8: The panel where additional modelling scenarios may be created. Enter a name in the "Name of Scenario" field, and change parameters from their baseline values using the parameter tabs. Click "+ Add Scenario" to add an additional scenario to the simulation; note that there may only be up to 4 additional scenarios at once.

On creating a new scenario, you will be presented with the panel seen in Figure 8. Type a name in the "Name of Scenario" field and press Enter to give the scenario a name. This name will be used on all graphs and tables generated from the output of simulation experiments that include the scenario. The panel may also be used to configure the new scenario's parameters separately from the baseline scenario. By default, the new scenario's parameters will be set to the values chosen for the baseline scenario, but they may be changed freely. You may also remove the scenario by clicking the "🗑 Remove Scenario" button below a scenario's name.

2 Running Simulations

To run your scenarios as part of a simulation experiment, use the sidebar to navigate to the "⏵ Run Simulations" page. This page contains several parameters which control the settings of the simulation engine, seen in Figure 9. These settings control aspects like how long the simulation will run for (in days), how many simulation runs to perform with each scenario (to account for how some parameters are stochastic) and what community the model will be based on. Simulation engine settings are applied to all scenarios rather than being individually configurable.

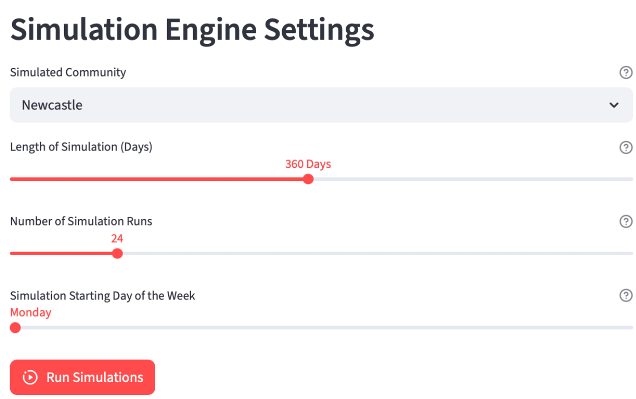

Figure 9: The "⏵ Run Simulations" page, with settings to configure the simulation engine. Click the "⏵ Run Simulations" button below them to run your scenarios together.

Once the simulation engine settings are configured, click the "⏵ Run Simulations" button below them. The confirmation prompt seen in Figure 10 will appear on the screen, listing the names of all configured scenarios. If there are any errors that result from the chosen parameter settings, such as an NPI being set to begin after the simulation has already ended, they will be listed beneath the parameter names (see Parameter Errors for more details). Click "Run

Simulation” at the bottom of the prompt to run the simulation experiment.

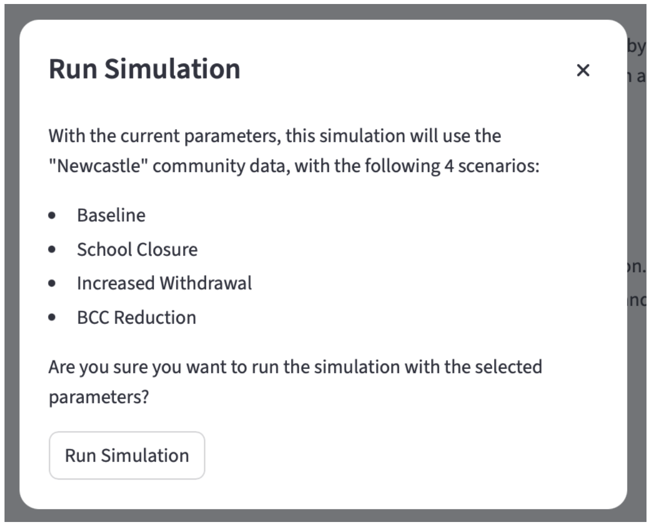

Figure 10: An example of the confirmation prompt that will appear when you attempt to run the simulation. In this instance, the user has added 3 more scenarios, naming them “School Closure”, “Increased Withdrawal” and “BCC Reduction”. Click “Run Simulation” at the bottom to send the parameters to the server.

When you run a simulation experiment, the dashboard sends the parameters for your scenarios to a separate server. This server will use these parameters to run the infectious disease model using the simulation engine. Due to the use of stochastic parameters, a single simulation experiment will involve multiple runs of the simulation engine for each scenario. Once all simulations are complete, the server will use the outcomes of each simulation to calculate the median daily infection counts for each scenario, then send these results back to the dashboard. While this process is occurring, you will not be able to run any new scenarios, but you can still navigate to other pages on the dashboard.

The simulation may take up to 30 minutes to run depending on how many scenarios were created. When the simulation is complete, a popup like the one in Figure 11 will appear in the top-right corner of the webpage to inform you that the results are ready.

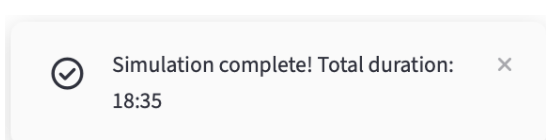

Figure 11: The popup that appears when the simulation is complete, which also lists how long it took to run.

3 Visualising Simulation Results

Once a simulation experiment has finished running and the server has returned its results, you will be able to use these results to generate graphs and tables.

3.1 Infection-Over-Time Graphs

The simplest of the visualisations the dashboard can create is a line graph plotting median infections over time (averaged over the multiple simulation runs in each experiment), such as the classical epidemic curves displayed in Figure 12. Use the sidebar to navigate to the “Infection Over Time Graphs” page in order to generate these graphs.

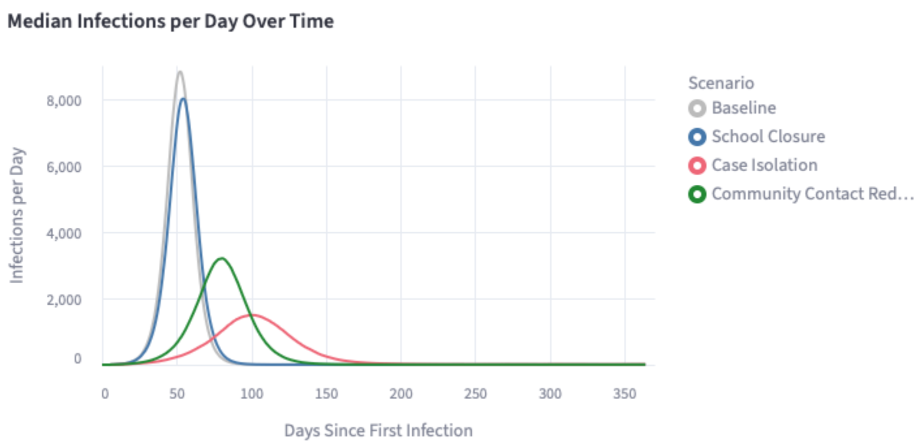

Figure 12: An example of an infection-over-time graph that can be generated using the dashboard, plotting daily infection curves under alternative mitigation measures.

Clicking on the “Graph Settings” menu seen in Figure 13 lets you configure whether the graph should display daily infections or total (cumulative) infections. It also contains a multiple selection box controlling which scenarios are displayed on the graph; by default, all scenarios defined in the “Scenario Parameters” page are included. Click the × next to a scenario to remove it, or click the ⊗ on the right to remove all of them; click the box itself to display any scenarios that can be added.

Click the “Create Graph” button to generate the graph. If the “Create Graph” button is greyed out, it may mean that your simulation hasn’t finished running yet; wait until it is complete before trying to generate a graph.

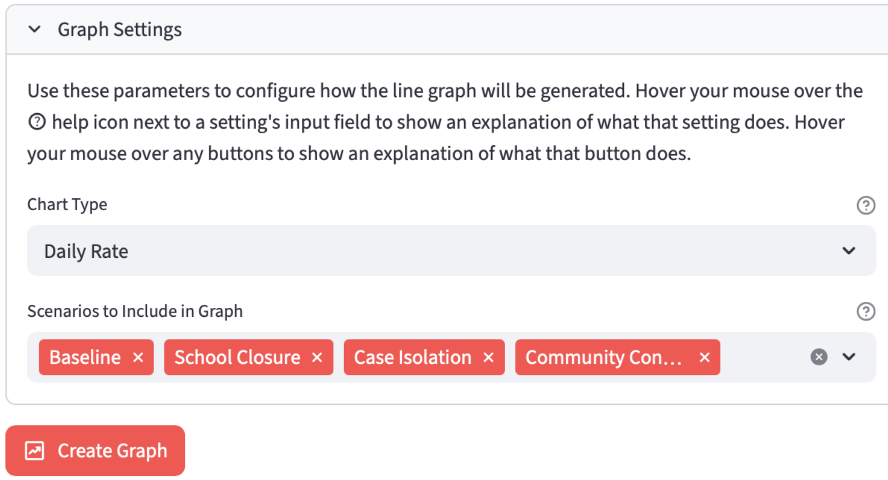

Figure 13: The settings panel for infection-over-time graphs, allowing you to choose the type of graph and which scenarios to include. Click the "Create Graph" button to use these settings to generate a graph.

Once the graph has been generated, it is displayed beneath the "Graph Settings" menu. This median daily infection graph can be interacted with in multiple ways:

- Hovering your mouse over the graph will display the infection counts for each scenario at that time-point on the graph.
- Clicking on the name of a scenario to the right of the graph will hide the lines for all scenarios except that one. Hold Shift while clicking on another scenario name to make that scenario's line reappear.
- Hovering your mouse over the top-right corner of the graph will display additional buttons, as seen in Figure 14:

Figure 14: The additional buttons that appear when your mouse is hovered over the top-right corner of a graph.

– Clicking the fullscreen icon will expand the image to fill the entire webpage. To return the image to normal size, click the exit fullscreen icon.

– Clicking the menu dots will display more options, including options to download the graph as a PNG or SVG image.

– Clicking the chart icon will display a spreadsheet containing the data used to generate the graph. To return to displaying the graph, click the graph icon that replaces the chart icon in the top-right corner.

If you wish to download a copy of the data used to create the graph, click the " Download Infection Data" button below the graph.

3.2 Health Burden Tables

The dashboard may be used to create outcome tables from the simulation experiment's generated data, compiling health burden outcomes as in Figure 15. These tables display various health burden outcomes, such as diagnosed cases, hospitalisations and deaths. The figures for these tables are made by combining the infection rates extracted from the simulation output with probabilities for each health burden outcome that are defined in the Disease tab when configuring the parameter settings for the experiment. Navigate to the " Health Burden Tables" page using the sidebar to create these tables.

Scenario	Age Group	Infections	Deaths (Difference from Baseline)
School Closure	Infant (7-24 Months)	4044	-6
School Closure	Child (6-12 Years)	11693	-38
School Closure	Adult (25-44 Years)	50280	-70
Case Isolation	Infant (7-24 Months)	1990	-109
Case Isolation	Child (6-12 Years)	5932	-327
Case Isolation	Adult (25-44 Years)	24942	-1337
Community Contact Reduction	Infant (7-24 Months)	2595	-78
Community Contact Reduction	Child (6-12 Years)	8216	-212
Community Contact Reduction	Adult (25-44 Years)	36543	-757

Figure 15: An example of a health burden table that can be generated using the dashboard, listing diagnosed cases between alternative mitigation methods for different age groups while highlighting how each method improves upon the baseline in terms of deaths.

Tables use a more advanced settings menu than graphs, as in Figure 16. The following settings can be configured here:

- Scenarios may be added or removed from the table, just as with infection-over-time graphs. Click the × beside each scenario to remove it, or click the ⊗ on the right to remove all of them. Click the selection box itself to display all the scenarios that can be added.
- Enabling the "Separate Results By Age Group" toggle will make the final table include separate rows for different age groups within each scenario. It also allows you to modify the "Age Groups to Include in Table" field below it.
- The "Age Groups to Include in Table" setting lets you choose which age groups to include in the table, including a Total group displaying the health burden outcomes for all age groups. The table in Figure 15 uses this setting to display health burden outcomes for 3 different age ranges.

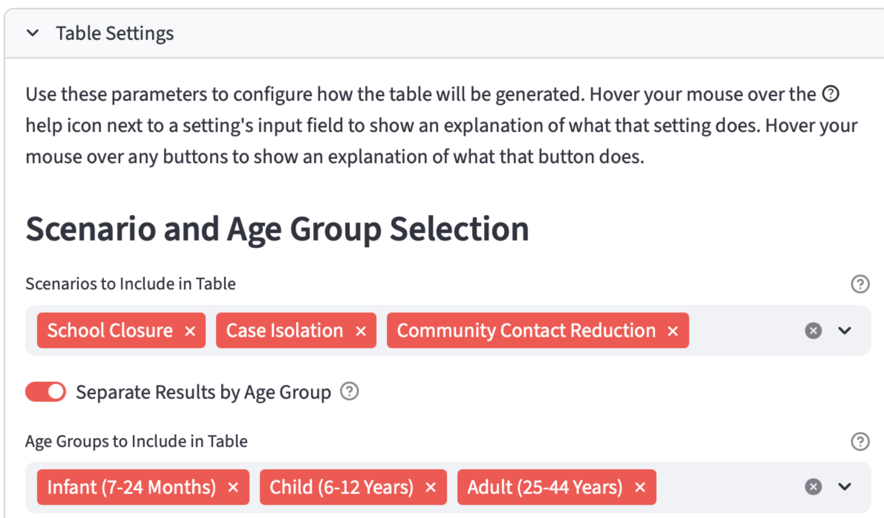

Figure 16: The settings panel for health burden tables, allowing you to configure which scenarios and age groups to include. The table where columns can be configured is not pictured here; see Figure 17 to view these column settings.

The columns that will be included in the table can be configured individually using the table seen in Figure 17, which works similarly to the tables used for age-specific parameters (see Tables). Double-click on a cell to begin editing it. Each row in the settings table represents a column that will be included in the final health burden table; rows can be added by clicking on the + icons on the bottom row of the table, or removed by clicking the ☐ checkbox on the left side of the table then the 🗑 delete icon at the top-right. Note that you may need to hover your mouse over the table to see these icons.

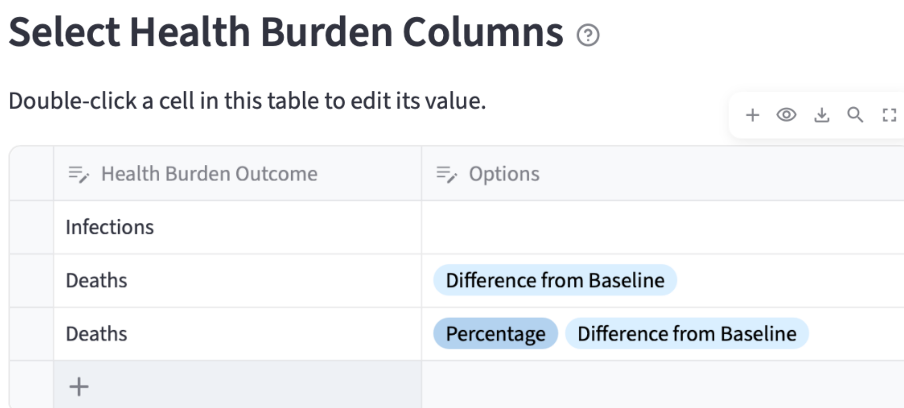

Figure 17: The column selection form for health burden tables. 3 columns have been configured: a column displaying total infections for each scenario, a column displaying the difference in total deaths for each scenario compared to the baseline scenario, and a column displaying the percentage difference in deaths from the baseline for each scenario.

The left column lets you select which health burden the column should display. These health burdens are calculated using the rates defined under the “Health Burden Outcomes” section of the ⚙ Disease parameters (see Configuring Simulations); this allows different scenarios to model diseases with different levels of severity. The right column lets you select options that modify how the columns are formatted. This column allows for multiple selections per row; you may select as many options as you like for each column, or none at all. The options that are available for each column are as follows:

- The “Difference from Baseline” option makes the column display the difference in health burdens between a given scenario and the baseline scenario, rather than displaying the raw values.
- The “Percentage” option makes the column display the percentage of the population instead of the raw values.
- If both “Difference from Baseline” and “Percentage” are selected, the column will display the percentage difference in health burdens between a given scenario and the baseline scenario.

Once you have selected your chosen settings, click the " Create Table" button to generate the table. Like the " Create Graph" button, the " Create Table" button may be greyed out if you have yet to complete a simulation or if no columns have been configured.

Like infection-over-time graphs, health burden tables are interactive in various areas:

- If the generated table is too large for the dashboard to display at once, you can scroll the table both vertically and horizontally using the scroll bars that appear on the right and bottom edges.
- The Scenario and Age Group columns are colour-coded for each category. Additionally, any "Difference from Baseline" columns are colour-coded based on the size of the difference; greater decreases in health burden outcomes are more blue, while greater increases are more red.
- The cells in the table are automatically displayed with scientific notation if their value is too large, while percentage values being rounded to 3 decimal places. Double-clicking on a cell in the table will display its exact value without this formatting.
- Clicking on a column header will sort the table using that column, in ascending order. Clicking again will switch to descending order, and clicking a third time will remove the sorting.
- Clicking and dragging a column header allows you to reorder columns.
- Clicking and dragging the borders between columns allows you to adjust how wide each column is.
- Hovering your mouse over a column header will display a ⋮ menu button. Clicking this button displays a menu where settings such as the column's number display format can be changed manually.
- Like the infection-over-time graphs, hovering your mouse over the top-right corner of a table will display additional buttons, as seen in Figure 18:

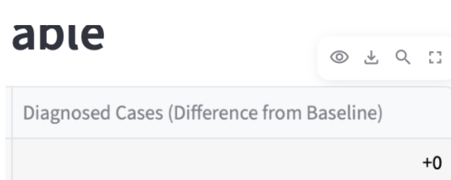

Figure 18: The additional buttons that appear when your mouse is hovered over the top-right corner of a table.

- Clicking the eye icon will allow you to remove columns from the table or add them back.
- Clicking the download icon will download a CSV copy of the table. The " Download Table Data" button below the table will also download the table; however, the icon in the top-right will account for any changes made to the table after it was generated, while the button will download a copy of the table as it was first generated.
- Clicking the search icon will allow you to find specific cells in the table by typing their values and pressing Enter.
- Clicking the fullscreen icon will expand the image to fill the entire webpage. To return the image to normal size, click the exit fullscreen icon.

4 Additional Notes

This section contains notes on what to do if errors occur alongside other features of the dashboard.

4.1 Appearance Settings

Clicking the menu dots at the top-right corner of the screen and selecting "Settings" from the options that appear will allow you to modify the dashboard's appearance. From here, you may toggle wide mode (stretching the contents of the dashboard to reach the edge of the screen) or switch between light and dark website themes.

4.2 Simulation Errors

If the dashboard cannot connect to the server, you will see the popup in Figure 19 appear when attempting to run the simulation. Ensure you are connected to the internet and try again. Other issues with the simulation will cause similar popups to appear; the error message will explain what to do.

4.3 Parameter Errors

If you set a parameter to a value that would cause issues in the simulation, an error message will appear above that parameter. The error message will explain

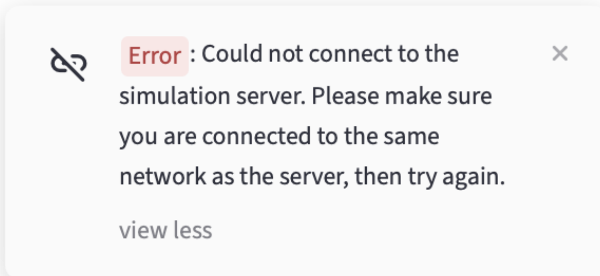

Figure 19: The popup that appears if the dashboard cannot connect to the server and run the simulation.

why the current values cause issues, and give you multiple options for how you can fix the issue. The error message will also appear at the top of the parameter selection page to ensure it is not overlooked.

Error messages may be either yellow or red. Yellow error messages are warnings; they appear if parameters will cause unrealistic or unintuitive behaviour. You may still run a simulation if your parameters receive yellow error messages, but you should read them carefully as they are likely to result from an oversight when setting parameters. Red error messages are crucial errors; the simulation will not run if any of these are present. Figure 20 shows examples of both red and yellow errors for the same parameter.

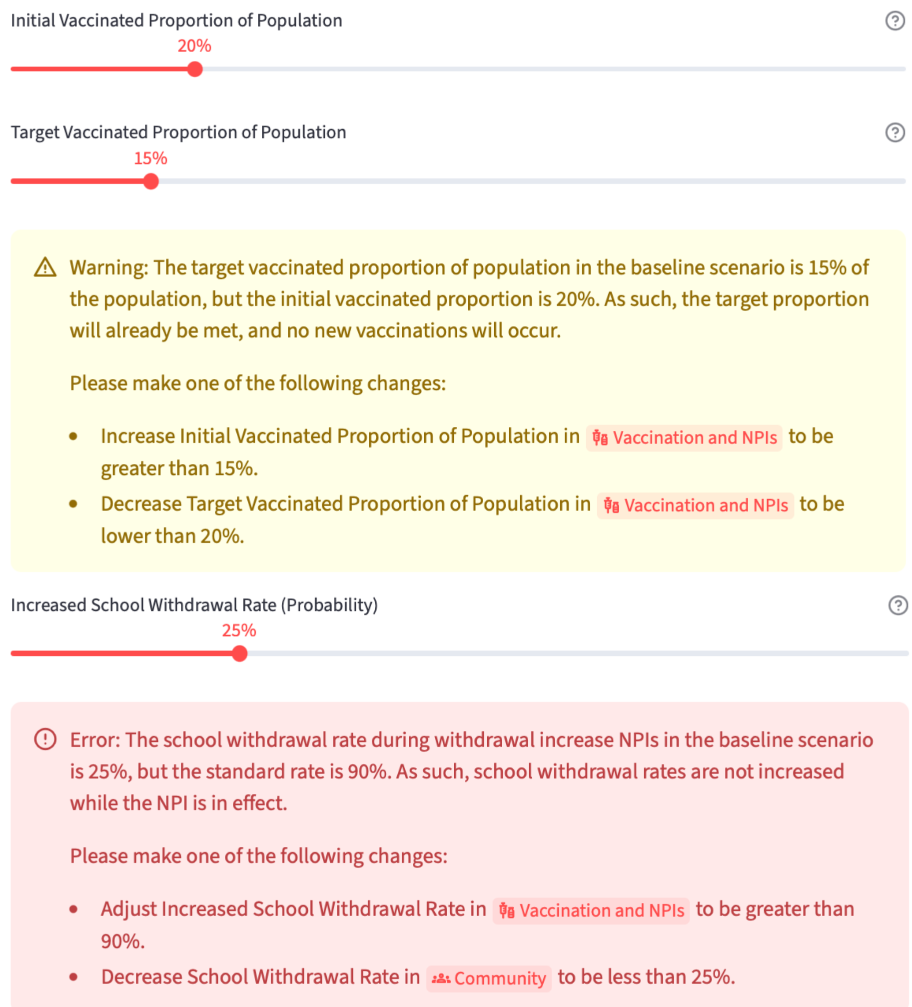

Figure 20: The two different types of error message that can appear next to parameters. Yellow messages are warnings, while red messages are crucial errors.

Glossary

***Flusim* model** The agent-based simulation model used by this dashboard to simulate infectious respiratory diseases. 19

background contact count A parameter defined as part of the *Flusim* model, which determines how many interactions between random people in the simulation occur each day. It is used to emulate interactions that occur outside of the model's simulated locations, such as on public transport or in sporting events. 5, 9

baseline scenario The scenario defined in the " Baseline Parameters" page. All other scenarios within a simulation experiment will use the parameters defined in this scenario as default values for parameters that the scenarios do not have their own values for. 6, 7, 12, 14

health burden outcome A statistic measuring a negative consequence of an infectious disease outbreak. Examples include GP visits, hospitalisations and deaths. 12, 19

non-pharmaceutical intervention Any intervention or protocol implemented to combat the spread of disease that does not involve medication or pharmaceutical products. Examples include social distancing and school closure. 3–5, 8

scenario A set of parameters defining a single environment for the simulation. 6–11, 13, 14, 19

simulation engine The program that conducts simulations according to the *Flusim* model, hosted separately from the dashboard itself. 8, 19

simulation experiment The process that is triggered upon clicking the " Run Simulations" button, in which multiple scenarios are ran together by the simulation engine in order to compare infection rates and health burden outcomes between them. 8, 10, 12, 19

simulation run The act of using the simulation engine to simulate a given scenario a single time. Simulation experiments use multiple runs for each scenario to account for the presence of stochastic parameters. 8, 10

stochasticity The property of being random or unpredictable. Several parameters used by the *Flusim* model are stochastic in order to replicate the unpredictable nature of infectious diseases. 8, 9, 19